

Einstein's Special Theory of Relativity



MD. GULAM MUSTAFA

Lecturer (Physics)

Department of Science & Humanities

Bangladesh Army International University of Science and Technology (BAIUST)

Syedpur, Cumilla-3501

Whatever we will discuss

➤ Definitions:

- a) Relativity
- b) Relativity theory
- c) Frame of reference
- d) Inertial frame of reference
- e) Non-inertial frame of reference

➤ State and explain the postulates of special theory of relativity.

➤ Explanation:

- a) Time dilation
- b) Proper time
- c) Length contraction
- d) Relativity of mass

Definitions

Absolute

Relative

Absolute

- Which will give same values for all observers

Relative

- Which will give different values for different observers

Definitions

Relativity

- According to classical mechanics space, time and mass are constant/absolute. Einstein said that space, time and mass are not absolute, these are relative. i.e., these give different values for different observers of different reference frames. This phenomenon is known as Relativity.

Theory of Relativity

- According to classical mechanics space, time and mass are constant/absolute. Einstein said that space, time and mass are not absolute, these are relative. This theory of Einstein is, therefore, called theory of relativity.

Frame of Reference

Frame of reference

- A frame of reference is a specified co-ordinate system in a three dimensional space with respect to which the motion of an object is described.

Inertial frame of reference

- The frames moving at constant speed with respect to each other and in which Newton's first law can be achieved are called inertial frame of reference.

Non-inertial frame of reference

- The frames having acceleration with respect to each other and in which Newton's first law cannot be achieved are called non-inertial frame of reference.

Postulates of Special Relativity

First postulates

- The laws of physics have the same form in all inertial frames of reference.

Second postulates

- The speed of light in free space has the same value in all inertial frames of reference.

Time Dilation

Time Dilation

- The time interval between the occurrence of two events which are in motion with respect to an observer will be longer than the time interval of occurrence of the same two events which are at rest with respect to the observer. This is called time dilation.

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

If $v = 0.98c$ then $t = 5t_0$

If $v = 0.999c$ then $t = 22.3t_0$

Related Numerical Problem

Problem-1: At the age of 25 years an astronaut went to space travel in a spaceship moving with velocity $1.8 \times 10^8 \text{ ms}^{-1}$ and he returns to earth after 30 years according to the calendar of the earth. What is his age now?

Ans : 49 years

Science fiction: Once a 40 years old scientist fell in love with his 16 years laboratory assistant. They want to marry; but they feel that their marriage cannot be welcomed by the society due to the age difference. Scientist plans to marry using the principle of time dilation of relativity. He goes to a long space journey in a spaceship moving with velocity (v) is equal to $0.999c$ [c = speed of light] for one year. He returns back and then their age difference barrier has been overcome, so they marry reciting the limerick.

Problem: What is the age difference between scientist & lab. assistant after the scientist returns back?

Ans : 2.63 years

Length Contraction

Length Contraction

- The length of an object in motion with respect to an observer appears to be shorter than when it is at rest with respect to him, this phenomenon is known as length contraction.

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Relativity of Mass

Relativity of Mass

- A body in motion with respect to the observer then its mass in motion will be greater than the mass of the body at rest with respect to him, this phenomenon is known as relativity of mass.

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$



Thank You
Thanks for your attention